18 At time $t = 0$ hours a tank is full of water.

Water leaks from the tank.
At the end of every hour there is 2% less water in the tank than at the start of the hour.

The volume of water, in litres, in the tank at time t hours is V_t

Given that

$$V_0 = 2000$$
$$V_{t+1} = kV_t$$

write down the value of k.

100% – 2% = 98% remains.

Multiplier = 98 ÷ 100 = 0.98.

	P	Factor	
×0.98	100	1	×0.98
	98	0.98	

$k =$ 0.98

1 mark
Use the proportion remaining

(Total for Question 18 is 1 mark)